

trailhead-agent

Engineering reference (CLI, pipeline, recording)

Version 0.5.1

1. Scope

- **IN:** Playwright discovery of Trailhead /content/learn/ unit links; LLM ordering under user intent (allowlisted hrefs); optional ranked URL walk; org checklist/prepare; .webm + JSON artifacts.
- **OUT:** Quiz submission, auto Launch Playground, arbitrary non-Trailhead crawling.

2. Runtime stack

```
Python 3.11+
playwright (Chromium)
langgraph + langchain-core (ranking graph)
pydantic (LLM JSON validation)
openai SDK (+ optional anthropic / google-genai extras)
```

3. Entrypoints

Console: trailhead-agent (trailhead_agent.cli:entrypoint). Module: python -m trailhead_agent <subcommand>

```
doctor
plan [--artifacts-dir|--save-e2e] [--walk-ranked N] [--json] ...
open-unit [--visit-count N] ...
auth
org doctor | org checklist | org prepare
```

4. CLI matrix

doctor	env/keys, executor, sf on PATH, recording dir hint
plan	discovery + LangGraph rank + optional walk; e2e JSON + .webm
open-unit	same rank path; goto first N ranked units only
auth	persistent Chromium profile for SSO
org doctor	sf org display JSON summary
org checklist	Markdown/JSON steps from plan JSON or unit href
org prepare	noop/cli executor checks; optional --open-playground browser

5. plan / open-unit shared flags

--config PATH	YAML browser/selectors
--artifacts-dir DIR	mkdir; sets TRAILHEAD_RECORD_VIDEO_DIR
--save-e2e	shorthand artifacts/e2e
--no-recording-demo-browser	-> TRAILHEAD_RECORDING_DEMO_AUTO=0
--start-url / --intent	or TRAILHEAD_* env

6. Data flow: plan while recording

```
1. demo_plan_what_you_asked (titlecard P1)
2. collect_units_for_seed(page, start_url)
```

```
3. demo_after_discovery (titlecard P2)
4. select_and_rank_units -> run_ranking_graph
5. demo_llm_ranking_output_slide (titlecard P3)
6. demo_after_llm_rank (titlecard P4)
7. walk_ranked_units_for_recording if --walk-ranked > 0 (titlecard P5)
8. _plan_recording_epilogue (live Trailhead + end card)
```

6b. Title cards

plan: P1 inputs | P2 discovery | P3 LLM ordered list | P4 next steps | P5 walk | end card
org prepare: 5 slides then module URL + corner tag + end card

7. LangGraph ranking

START -> prepare -> planner (optional LLM_PLANNER_PHASE=1)
-> rank_primary -> rank_repair OR finalize -> END

8. Key modules

cli.py	argparse, trace_id, artifacts-dir bootstrap
runner.py	run_dry_plan, open_first_ranked_unit, walk, epilogue
discovery.py	unit links, DiscoveryError
llm_agent.py	providers, select_and_rank_units
ranking_graph.py	LangGraph compile + run_ranking_graph
session.py	TrailheadBrowser, Playwright video
e2e_artifacts.py	JSON, friendly .webm rename, manifest
e2e_viewer.py	e2e-report.html
demo_titlecards.py	data-URL slides
demo_narration.py	stderr narration
demo_browser.py	headed/slow_mo when recording
org_commands.py	prepare / checklist / doctor

9. Artifact files

plan-primary.webm | open-unit-primary.webm | org-prepare.webm
e2e-plan-<trace>.json, e2e-plan-latest.json
e2e-open-unit-latest.json
e2e-org-prepare-latest.json, e2e-manifest.json, e2e-report.html

10. Environment variables

LLM_PROVIDER, OPENAI_* | GOOGLE_* / GEMINI_* | ANTHROPIC_*
LLM_PLANNER_PHASE, LLM_RANKING_REPAIR
TRAILHEAD_START_URL, TRAILHEAD_INTENT
TRAILHEAD_RECORD_VIDEO_DIR, TRAILHEAD_RECORDING_DEMO_AUTO
TRAILHEAD_DEMO_TITLECARDS, TRAILHEAD_DEMO_TITLECARD_MS, TRAILHEAD_DEMO_TITLECARD_RANK_MAX
TRAILHEAD_DEMO_NARRATION, TRAILHEAD_DEMO_SLOW_MO_MS, TRAILHEAD_AGENT_HEADLESS
TRAILHEAD_PLAN_RECORDING_TAIL_MIN_MS, TRAILHEAD_ORG_PREPARE_RECORDING_MIN_MS
TRAILHEAD_PAGE_SETTLE_MS, TRAILHEAD_DISCOVERY_*, TRAILHEAD_SHELL_WAIT_TIMEOUT_MS
TRAILHEAD_ORG_EXECUTOR, TRAILHEAD_SF_ORG_ALIAS

11. Errors / exit codes

DiscoveryError	scrape failure; details + hints
UrlValidationError	URL policy
LLMProviderError	API/schema
ConfigurationError	YAML/env
OrgExecutorError	org/sf integration
KeyboardInterrupt	130

12. Verification

```
pytest tests
(TRAILHEAD_DEMO_NARRATION=0 in tests/conftest.py)
```

13. PDF generation

Preferred: open this HTML file in Chrome/Edge, **Ctrl+P, Save as PDF**. See `HOW_TO_SAVE_PDF.md`.

Optional: `pip install -e "[docs]"` then `python scripts/generate_engineering_pdf.py` (fpdf2; some viewers render poorly).

14. Demo recording contract

Example: module `apex_database`, intent `Hands-on Apex only`, `--walk-ranked 2`; org prepare with same `start_url` and `TRAILHEAD_RECORD_VIDEO_DIR` on the artifacts directory.

Ctrl+P → Save as PDF